MILITARY SPECIFICATION

TRANSISTOR, TYPE 2N463

1. SCOPE

1.1 Scope. - This specification covers the detail requirements for germanium, PNP, power transistors, the general requirements which are in accordance with Specification MIL-S-19500, unless otherwise specified herein. The tranistors have the following salient characteristics at a case temperature of 25°± 3°C:

	h_{FE} V_{CE} = -2 Vdc I_C = -2 Adc	h_{FE} V_{CE} = -5 Vdc I_C = -0.5 Adc	h_{IE} V_{CE} = -2 Vdc I_C = -2 Adc	g_{ME} V_{CE} = -2 Vdc I_C = -2 Adc
Min.	20	30	4 ohms	2 mhos
Max.	60	120	15 ohms	6 mhos

	$V_{CE(sat)}$ I_C = -2 Adc I_B = -0.2 Adc	f_{ae} V_{CE} = -5 Vdc I_C = -0.5 Adc	θ_{J-C}
Min.	---	5.0 kc	---
Max.	0.16 Vdc	---	1.5 °C/W

1.2 Absolute maximum ratings. -

P_C [1/] T_C=25°C.	T_{stg} °C.	V_{CEV} [2/] Vdc	V_{CBO} [2/] Vdc	V_{EBO} [2/] Vdc	I_B Adc	I_E Adc
W 50	-65 to +100	-60	-60	-50	-1	5

[1/] For collector power ratings at mounting surface temperatures higher than 25°C, P_C is derated at 0.67 W/°C (within the range +25° to +100°C).

[2/] These voltage ratings may be exceeded (without permanently impairing the serviceability of the transistor) provided the power is limited to 0.5 watts.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

FSC 5960

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Transistors shall be in accordance with Specification MIL-S-19500 and as specified herein.

3.2 Design and construction.- Transistors shall be of the design, construction and physical dimensions shown on figure 1.

3.3 Abbreviations and symbols.- The following abbreviations and symbols shall also apply:

BV_{CEV} . . . . Collector to emitter breakdown voltage, reverse bias.

V_{CEV} . . . . Collector to emitter voltage, reverse bias.

3.4 Performance characteristics. Performance characteristics shall be as specified in 4.5, 4.6 and 4.7.

3.5 Marking.- Transistors shall be marked with the manufacturer's identification, "USN" prefix, type designation, acceptance date, and lot identification suffix letter.

3.5.1 Lot identification suffix letter.- Each inspection lot shall be identified by a capital Gothic letter. The suffix letter shall be marked on each transistor immediately following the coded acceptance date. A different letter shall be used for each acceptance lot in any week.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.5, 4.6 and 4.7. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1)).

4.2 Delayed delivery.- Devices held for a period exceeding 6 months following source inspection acceptance shall be reinspected for all Group A requirements prior to shipment.

4.3 Inspection conditions.- Unless otherwise specified, electrical measurements shall be taken within 3 seconds of application of bias, except for collector and emitter cutoff currents and breakdown voltages which shall be taken at 7 ± 3 seconds of application of bias and the case temperature shall be maintained at 25 ± 3°C.

4.4 Acceptance inspection.- Acceptance inspection shall consist of examinations and tests specified in 4.5, 4.6 and 4.7. Sampling shall be in accordance with 4.4.1 except for life tests which shall be in accordance with Method B of Specification MIL-S-19500.

4.4.1 Acceptance sampling.- The acceptance sample size shall be chosen by the manufacturer and shall be adequate to assure, with a 90 percent confidence, the specified Lot Tolerance Percent Defective (LTPD). At the discretion of the manufacturer, 100 percent testing of the inspection lot to the specified LTPD shall be allowed.

4.4.2 Acceptance number.- The acceptance number associated with the selected sample size shall be determined using a suitable acceptance sampling plan which will assure the specified LTPD with a 90 percent confidence.

4.4.3 Table I shows the minimum sample size necessary to assure a maximum LTPD, with a confidence level of 90 percent for various failure acceptance numbers (a). (Rejection number, r, equals a + 1).

4.4.4 Selection of samples.- The selection of a sample size shall be at the option of the manufacturer. The minimum quality (actual percent defective of the manufactured lot necessary to accept 95 percent (19 out of 20) of the lots submitted is shown for information in table II. The values shown are equivalent to AOL's for the respective sample sizes and acceptance numbers shown in table I.

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4.4.5 Additional samples.- The manufacturer may add an additional quantity to the initial sample, but this may be done only once for any subgroup and the added samples shall be subjected to all the tests within the subgroup. The total sample (initial and added samples) shall determine the new acceptance number. The total defectives of the initial and second sample shall be additive and shall comply with the specified LTPD requirement.

4.4.6 Tightened inspection.- Tightened inspection on resubmitted lots is obtained by testing to an LTPD equal to the next lower number in the series 1, 1.5, 2, 3, 5 and 7 x 10^n from the specified initial LTPD.

4.4.7 Reduced inspection.- Eligibility for reduced inspection shall be achieved when the following criteria have been met:

(a) The immediately preceding five lots have met the acceptance requirements.
(b) The average percent defective over the preceding five lots has not exceeded one-tenth of the specified LTPD.
(c) There has been no unusual discontinuity in production in the immediately preceding five lots.

4.4.7.1 The manufacturer shall lose eligibility for reduced inspection whenever two of five consecutive lots have failed the acceptance requirements or the percentage defective over the preceding five lots exceeds two-tenths of the specified LTPD. (In case of a single lot failure, the sample for inspection shall have been completely tested to determine the total number of defectives for percentage defective calculation.)

4.4.7.2 Reduced inspection is performed by testing to an LTPD equal to the next higher number in the series 1, 1.5, 2, 3, 5 and 7 x 10^n from the specified initial LTPD.

4.4.8 Acceptance-inspection information.- One copy of the acceptance-inspection information pertinent to the transistor inspection lot shall be furnished by the transistor supplier and shall accompany each transistor shipment from the inspection lot to the equipment manufacturer.

4.5 Group A inspection.- Group A inspection shall consist of the examinations and tests specified in table III. A device having one or more defects shall be counted as one defective.

4.6 Group B inspection.- Group B inspection shall consist of the examinations and tests specified in table IV.

4.6.1 Disposition of sample units.- Sample units which have been subjected to and have passed subgroups 1, 2 and 5 of group B inspection and subgroup 2 of Group C inspection may be delivered on the contract or order provided that these sample units are subjected to and pass Group A inspection. Units found defective in the course of inspection shall not be delivered on the contract or order.

4.7 Group C inspection.- Group C inspection shall consist of the examinations and tests specified in table V.

4.7.1 Salt atmosphere.- The device shall be examined for destructive corrosion and illegible marking.

4.8 Samples for destructive tests.- Sample devices for the tests in subgroups 3 and 4 of group B inspection and subgroup 1 of group C inspection may be units which are outside Group A requirements by less than 10 percent on the following parameters: g_{ME}, $V_{CE}(sat)$, h_{FE}(2 Adc), h_{FE}(0.5 Adc), h_{ie}, $f_{\alpha e}$.

5. PREPARATION FOR DELIVERY

5.1 See Specification MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.).

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Table I - Minimum size of sample to be tested to assure, with a 90 percent confidence, a lot percent defective no greater than the LTPD specified.

Maximum percent defective (LTPD)	20	15	10	7	5	3	2	1.5	1	0.7
Acceptance number (a) (r = a + 1)	Minimum sample sizes									
0	11	15	22	32	45	76	116	153	231	328
1	18	25	38	55	77	129	195	258	390	555
2	25	34	52	75	105	176	266	354	533	759
3	32	43	65	94	132	221	333	444	668	953
4	38	52	78	113	158	265	398	531	798	1140
5	45	60	91	131	184	308	462	617	927	1323
6	51	68	104	149	209	349	528	700	1054	1503
7	57	77	116	166	234	390	589	783	1178	1680
8	63	85	128	184	258	431	648	864	1300	1854
9	69	93	140	201	282	471	709	945	1421	2027
10	75	100	152	218	306	511	770	1025	1541	2199

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Table II - Minimum quality required to accept (on the average) 19 of 20 lots.

Maximum percent defective (LTPD)	20	15	10	7	5	3	2	1.5	1	0.7
Acceptance number (a) (r = a + 1)	Minimum quality (percent defective) (equivalent to AQL)									
0	0.46	0.34	0.23	0.16	0.11	0.07	0.04	0.03	0.02	0.02
1	2.0	1.4	.94	.65	.46	.28	.18	.14	.09	.06
2	3.4	2.4	1.6	1.1	.78	.47	.31	.23	.15	.11
3	4.4	3.2	2.1	1.5	1.0	.62	.41	.31	.20	.14
4	5.3	3.9	2.6	1.8	1.3	.75	.50	.37	.25	.17
5	6.0	4.4	2.9	2.0	1.4	.85	.57	.42	.28	.20
6	6.6	4.9	3.2	2.2	1.6	.94	.62	.47	.31	.22
7	7.2	5.3	3.5	2.4	1.7	1.0	.67	.51	.34	.24
8	7.7	5.6	3.7	2.6	1.8	1.1	.72	.54	.36	.25
9	8.1	6.0	3.9	2.7	1.9	1.2	.77	.58	.38	.27
10	8.4	6.3	4.1	2.9	2.0	1.2	.80	.60	.40	.28

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TABLE III - Group A inspection

Examination or test	Conditions	LTPD	Symbol	Limits		Units
				Min.	Max.	
Subgroup 1						
Visual and mechanical examination	---	5	---	---	---	---
Subgroup 2						
Collector cutoff current, open emitter	V_{CB} = -40 Vdc	3	I_{CBO}	---	-300	µAdc
	V_{CB} = -2 Vdc		I_{CBO}	---	-100	µAdc
Emitter cutoff current, open collector	V_{EB} = -40 Vdc		I_{EBO}	---	-500	µAdc
	V_{EB} = -2 Vdc		I_{EBO}	---	-100	µAdc
Emitter to base breakdown voltage, open collector	I_E = -3 mAdc		BV_{EBO}	-50	---	Vdc
Collector to emitter breakdown voltage, reverse voltage bias base	I_C = -3 mAdc V_{BE} = .75 Vdc		BV_{CEV}	-60	---	Vdc
Subgroup 3						
Static transconductance	V_{CE} = -2 Vdc I_C = -2 Adc	3	g_{ME}	2	6	mhos
Saturation voltage (collector to emitter)	I_C = -2 Adc I_B = -0.2 Adc		$V_{CE}(sat)$	---	-0.16	Vdc
Static forward current transfer ratio	V_{CE} = -2 Vdc I_C = -2 Adc		h_{FE}	20	60	---
	V_{CE} = -5 Vdc I_C = -0.5 Adc		h_{FE}	30	120	---
Static input impedance	V_{CE} = -2 Vdc I_C = -2 Adc		h_{IE}	4	15	ohms
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	V_{CE} = -5 Vdc I_C = -0.5 Adc		$f_{\alpha e}$	5.0	---	kc

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TABLE IV - Group B inspection

Examination or Test	Conditions	LTPD	Symbol	Limits		Units
				Min.	Max.	
Subgroup 1						
Physical dimensions	---	15	---	---	---	---
Subgroup 2						
Thermal resistance	---	10	θ_{J-C}	---	1.5	°C/W
High temperature operation:	T_C = 70°C.					
Collector cutoff current, open emitter	V_{CB} = -2 Vdc		I_{CBO}	---	-2	mA
Low temperature operation:	T_C = -40°C					
Static forward current transfer ratio	I_C = -2 Adc V_{CE} = -2 Vdc		h_{FE}	15	---	---
Subgroup 3						
Soldering	---	10	---	---	---	---
Temperature cycling	5 cycles $T_{(high)}$= 100° $^{+0°}_{-5°}$C.		---	---	---	---
Thermal shock (glass strain)	$T_{(high)}$= 85° ± 5°C. & $T_{(low)}$= 0° ± 2°C.					
Moisture resistance	---		---	---	---	---
End points:						
Collector cutoff current, open emitter	V_{CB} = -40 Vdc		I_{CBO}	---	-500	µAdc
Emitter cutoff current, open collector	V_{EB} = -40 Vdc		I_{EBO}	---	-750	µAdc
Collector to emitter breakdown voltage, reverse voltage biased base	V_{BE} = .75 Vdc I_C = -5 mAdc		BV_{CEV}	-60	---	Vdc
Subgroup 4						
Shock	5 blows, 500 G X_1, Y_1 & Z_1		---	---	---	---
Constant acceleration	5,000 G X_1, Y_1 & Z_1					
Vibration fatigue	V = 0		---	---	---	---

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TABLE IV - Group B inspection (Cont'd.)

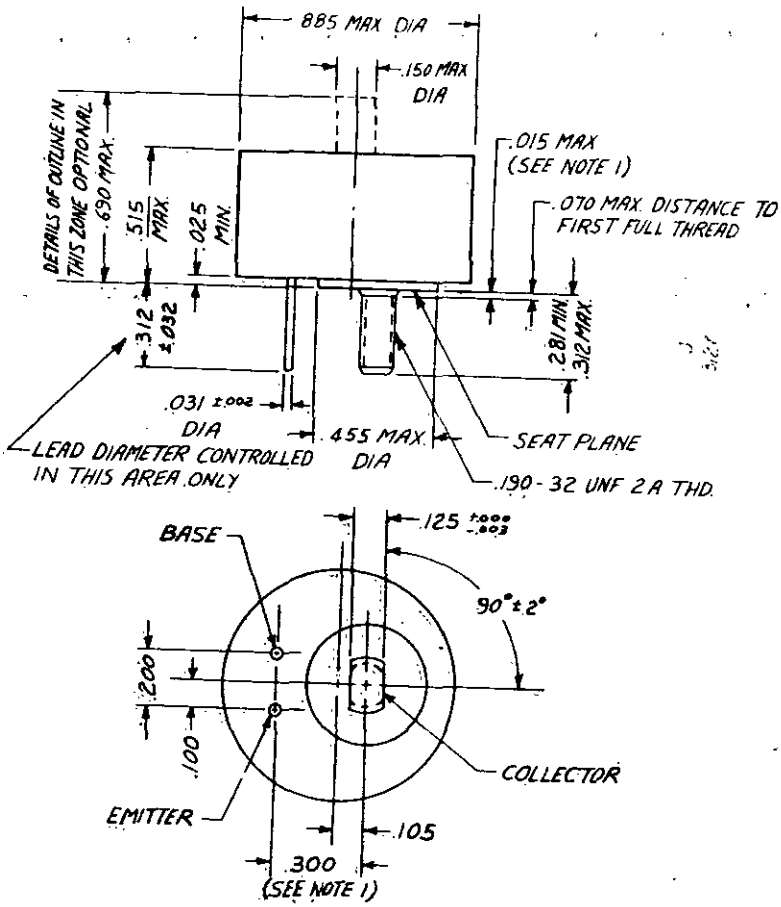
Examination or Test	Conditions	LTPD	Symbol	Limits		Units
				Min.	Max.	
Subgroup 4 (Cont'd.)						
Vibration, variable frequency	---	10	---	---	---	---
End points: (Same as for subgroup 3)						
Subgroup 5						
Storage life	Method B $T_A = 100° \ ^{+0}_{-5}°C.$	λ=10%	---	---	---	---
End points:						
Collector cutoff current, open emitter	V_{CB} = -40 Vdc		I_{CBO}	---	-600	µAdc
Emitter cutoff current, open collector	V_{EB} = -40 Vdc		I_{EBO}	---	-1.0	mAdc
Collector to emitter breakdown voltage, reverse voltage biased base	I_C = -6 mAdc V_{BE} = .75 Vdc		BV_{CEV}	-60	---	Vdc
Static forward current transfer ratio	V_{CE} = -2 Vdc I_C = -2 Adc		h_{FE}	15	---	---

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TABLE V - Group C inspection

Examination or Test	Conditions	LTPD	Symbol	Limits		Units
				Min.	Max.	
Subgroup 1						
Salt atmosphere (corrosion) End points: (Same as for subgroup 5 of table IV.)	Period = 6 months See 4.7.1	15	---	---	---	---
Subgroup 2						
Operation life End points: (Same as for subgroup 5 of table IV)	Period = 1 month Method B $T_C = (85 \pm^{0}_{5})°C.$ $P_C = (10 \pm^{0}_{1})$ W $I_E = (1 \pm 0.1)$Adc	λ=10	---	---	---	---

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NOTES:

1: Maximum size leads and stud shall be within 0.0055 of the exact locations shown with respect to a 0.885 diameter measured at points 0.015 maximum below seating plane.
2: All dimensions are in inches.

Figure 1 - Dimensions of transistor, type 2N463.

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